

KALINGA INSTITUTE OF INDUSTRIAL TECHNOLOGY

Deemed to be University, Bhubaneswar - 751024, Odisha

MID SEMESTER EXAMINATION — EXAMINATION QUESTION PAPER

Subject: DATABASE MANAGEMENT SYSTEMS

Time: 1.5 Hours

Semester / Programme: 4

Full Marks: 20

Instructions: Answer Question No. 1 (Compulsory) and any THREE from Question No. 2 to 5. Figures in brackets indicate marks.

SECTION - A (Compulsory)

- 1(a)** What is the primary difference between a physical schema and a logical schema? [1]
- 1(b)** Define the term 'super key' in the relational model. [1]
- 1(c)** Express the Armstrong's Axiom of Transitivity in symbolic notation. [1]
Formula: If $X \twoheadrightarrow Y$ and $Y \twoheadrightarrow Z$, then $X \twoheadrightarrow Z$
- 1(d)** What is the purpose of the 'NULL' value in SQL? [1]
- 1(e)** Identify the primary advantage of using a DBMS over a file processing system. [1]

SECTION - B (Answer any THREE)

Answer any THREE questions from Q.2 to Q.5. Each question carries 5 marks.

- Q.2. (a)** Explain the three-schema architecture of a database system. [2.5]
- (b)** Differentiate between Total Participation and Partial Participation in E-R modeling with an example. [2.5]
- Q.3. (a)** Given the relation $R(A, B, C, D)$ and functional dependencies $F = \{A \twoheadrightarrow B, B \twoheadrightarrow C, C \twoheadrightarrow D\}$. Find the closure of A (A^+). [2]
 A^+
- (b)** State and prove the Augmentation rule of Armstrong's Axioms. [3]
Formula: If $X \twoheadrightarrow Y$, then $XZ \twoheadrightarrow YZ$
- Q.4. (a)** Explain the concept of 'Lossless Decomposition' in database normalization. [2.5]
Formula: $R_1 \cap R_2 \twoheadrightarrow R_1$ or $R_1 \cap R_2 \twoheadrightarrow R_2$
- (b)** Briefly define 1NF and 2NF. [2.5]
- Q.5. (a)** Write a SQL query to find the names of employees who earn more than the average salary of their respective departments. [3]
- (b)** Explain the difference between 'WHERE' and 'HAVING' clauses in SQL. [2]

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